# Maintenance Schedule

Diesel engine

**20V956TB34**
**for emergency gensets in nuclear power stations only**
**according to IEEE 387 and KTA 3702**
**Application group 3C**

**MS50158/04E**

A Rolls-Royce solution

| Engine model | kW/cyl. | Application group |
|---|---|---|
| 20V956TB34 | 325 kW/cyl. | 3C, prime power limited, ICXN |
| Notice: Non-scheduled revision must be carried out in the event of: | - Incident acc. to IEEE or KTA<br>- Safety shutdown earthquake<br>- Operation beyond the admissible operating limits (operation beyond design base) | |
| Load profile: | Continuous power as per IEEE 387 and ISO 8528-1 | |

*Table 1: Applicability*

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# 1 Maintenance Schedule

## 1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

### Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

### Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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## 1.2 Special conditions

| Note for engines in standby applications: |
| --- |
| The maintenance concept of the operator must take into account the event / incident at which a standby power demand occurs and the maintenance intervals to be observed for all components. This means that maintenance tasks will be due earlier by the time the engine is running in the event of a standby power demand.<br>The maintenance schedule is based on normal engine operation as a standby genset engine with monthly test runs according to KTA/IEEE (the number of starts per month is assumed to be approx. 3) |

*Table 2: Special conditions*

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## 1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

| Interval (HRS)/limit | Limit in % | Limit max. |
|---|---|---|
| Operating hours | ± 10% | ± 500 hrs |
| Years | ± 10% | ± 30 days |
| Example 1 | Task every 1000 hrs | Between 900 hrs and 1100 hrs (10% of 1000 hrs) |
| Example 2 | Task every 30000 hrs | Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs) |

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## 1.4 Load profile

| Load | 110% | 100% | 85% | <15% | 5% |
| --- | --- | --- | --- | --- | --- |
| Corresponding operating time | <1% | 8.5% | 70% | 5.5% | 15% |

*Table 3: Load profile*